# Detecting Logic Discrepancies in Cross-Layer Communications within Mini-Program Frameworks

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## 1 Known Issues

Table 1. Known issues of logic discrepancies in cross-layer communications.

| Issue | Discrepancies | Data Channel | Data Type | Page Element | Page Position | Observation Oracle | Combination | UI Interaction | Super App (Version) | Reference |
|---|---|---|---|---|---|---|---|---|---|---|
| K1 | DEV : EW \| GS | ③ | array | * | * | data value monitor | none | none | WeChat (3.4.0) | [4] |
| K2 | Android : GW \| GS | ③ | array | * | content | data value monitor | none | none | WeChat (3.4.0) | [8] |
| K3 | Android : GW \| GS | ① | object | * | content | data value monitor | none | none | WeChat (3.4.3) | [7] |
| K4 | DEV : EW \| GS | ③ | string | * | * | data value monitor | none | none | WeChat (3.4.0) | [9] |
| K5 | iOS : EW \| GS | ③ | array | * | wx:for | data value monitor | none | none | WeChat (3.4.3) | [5] |
| K6 | DEV : EW \| GS | ③ | array | * | attribute | data value monitor | none | none | WeChat (3.0.2) | [1] |
| K7 | GW : Windows \| Android iOS | ⑨ | * | * | attribute | event object inspector | none | button tapping | WeChat (3.5.2) | [3] |
| K8 | DEV : EW \| GS | ⑨ | * | * | attribute | event object inspector | none | button tapping | WeChat (3.5.8) | [6] |
| K9 | GS : iOS \| DEV | ⑨ | * | * | attribute | event object inspector | none | button tapping | WeChat (3.4.10) | [10] |
| K10 | WR : DEV \| Android iOS | ⑨ | * | * | attribute | event object inspector | none | button tapping | Alipay (2.9.73) | [11] |
| K11 | DEV \| Android iOS | ① | array | * | a:for | data value monitor | none | none | TikTok (1.57.0) | [2] |

We collected 11 relevant reports(§6.2). The analysis results are presented in Table 1. We make the following observations. First, these known issues can be triggered by individual instances of cross-layer communication patterns, without requiring combinations of multiple instances. Second, these known issues are readily observable and do not require complex observation oracles.

In the `Discrepancies` column, the icon means platform: Android (Android), iOS (iOS), Windows (Windows), macOS (macOS), and DevTools (DEV) for WeChat. The letter means configuration: Exparser+WebView (EW), GlassEasel+Skyline (GS), GlassEasel+WebView (GW).

In the `Data Channel` column, the number (①-⑨) means the types of Data Channel: Channel ① (RJS $\overset{var}{\rightsquigarrow}$ XML), Channel ② (RJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ③ (RJS $\overset{upd}{\rightsquigarrow}$ XML), Channel ④ (RJS $\overset{upd}{\leftrightsquigarrow}$ XML), Channel ⑤ (EJS $\overset{var}{\rightsquigarrow}$ XML), Channel ⑥ (EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑦ (RJS $\overset{var}{\rightsquigarrow}$ EJS $\overset{fun}{\rightsquigarrow}$ XML), Channel ⑧ (XML $\overset{ev}{\rightsquigarrow}$ RJS), Channel ⑨ (XML $\overset{ev}{\leftrightsquigarrow}$ EJS).

2026. ACM XXXX-XXXX/2026/3-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

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